



Jorge Ramón
Parra Michel, Ph.D.

Mechanical Engineer | Optical
Metrology Researcher

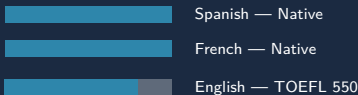
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NATIONALITIES

- Mexican | French

LANGUAGES



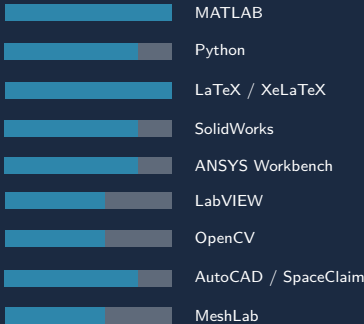
PROFILES

- 0000-0002-9145-5574
- Google Scholar
- Web of Science

PROFESSIONAL LI-
CENCES

Ph.D. Licence No. 00019609
M.Sc. Licence No. 00019608
Mech. Eng. Licence No. 5384881
ENIC-NARIC EU Attestation

SOFTWARE & TOOLS



Objective

To advance optical metrology and mechanical stress analysis through physics-based optical techniques and computational modeling. Expert in Electronic Speckle Pattern Interferometry (ESPI), Digital Image Correlation (DIC), and Structured Light Projection for 3D shape and deformation measurement. Skilled in Finite Element Analysis, precision manufacturing, and reverse engineering.

Professional Goals

To lead research in optical instrumentation, physics-informed deep learning for 3D topography reconstruction, and industrial metrology combining optics, mechanics, and AI. Strengthen the interface between scientific research, technological innovation, and higher education in engineering.

Fellowships & Recognition

2023–present	SNI Level 1 — SECIHTI <i>Mexican National System of Researchers</i>
2025	Beca Santander–FIMPES Investigación <i>Banco Santander Universidades & FIMPES México</i> Project: <i>Development and Assessment of Competencies in Advanced Manufacturing: AI Applications in Optical Inspection, Reverse Engineering, and Computational Analysis.</i>

Education

2007–2011	Ph.D. in Optics <i>Centro de Investigaciones en Óptica (CIO), México</i> Thesis: <i>In-Plane ESPI for mechanical stress analysis and topometry using divergent illumination.</i> ENIC-NARIC: Doctorat (Level 8 EQF).
2004–2006	M.Sc. in Optics <i>Centro de Investigaciones en Óptica (CIO), México</i> Thesis: <i>ESPI and phase shifting in dynamic events for analysis of uniaxial deformation in welds.</i> ENIC-NARIC: Master (Level 7 EQF, Ref. 2080168).
2000–2004	Mechanical–Electrical Engineer <i>Universidad de Guadalajara, México</i> ENIC-NARIC: Licence/Bachelor (Level 6 EQF, Ref. 821055).

RESEARCH LINES

- ESPI / Interferometry
- Structured Light 3D
- Digital Image Correlation
- Physics-Informed DL
- Reverse Engineering
- Mechanical Design & FEA

PATENTS

- Wind Generator for Forced Draft Pipes (MX/2015/001586A)
- Machine to Separate Solid & Light Particles from Air (MX/2016/000865A)

TEACHING (CURRENT)

LEM430 — Steel Structures
MDM410 — 3D Simulation & Prototyping
LIB427 — Microcontrollers
Research Methodology

Professional Experience

2007–present	Professor–Researcher <i>Universidad La Salle Bajío, A.C., León, México</i> • Research in optical metrology, 3D shape measurement, stress–strain analysis. • Design of interferometric and structured-light instrumentation. • Computational tools (MATLAB, Python) for optical data processing. • Teaching at undergraduate and graduate levels. • R&D grant proposals and industry collaboration.
2019–present	Independent Consultant — 3D Optical Measurement 3D scanning and CAD modeling for mechanical, artistic, and industrial components using laser and structured-light systems for additive manufacturing and CNC production.
1998–2004	Mechanical Engineer (Jr.) <i>Industria Metálica de León, C.V.</i> Design, drafting, and structural analysis of mechanical assemblies.

Selected Publications

2026	Parra-Michel, J.R. , Martínez Santamaría, N.A., Aguilar Vargas, E. <i>Solving the Phase Unwrapping Problem in Dual Divergent Projection Systems Using a Physics-Informed Neural Network.</i> Proc. ICECET 2026, Rome, Italy (Paper ID 1199, accepted).
2026	Parra-Michel, J.R. , Martínez Santamaría, N.A., Aguilar Vargas, E. <i>Physics-informed hybrid U-Net for phase-to-height reconstruction in dual divergent structured-light systems (PIHUNN).</i> Optics and Lasers in Engineering , 201, 109693. doi:10.1016/j.optlaseng.2026.109693
2022	Hernández, A.R.C., Ontiveros, R.I.R., Ponce, H.E.E., Ramírez, J.M.O., Parra-Michel, J.R. , Moller, J.A.D. <i>Effect of silver on structural, optical, and electrical properties of ZnO:Al/Ag/ZnO:Al thin films.</i> Nova Scientia , 14(29). doi:10.21640/ns.v14i29.3038
2021	Parra-Michel, J.R. , Martínez-Peláez, R., Duarte-Moller, J.A. <i>Double structured light with divergent projection for surface topometry.</i> Meas. Sci. Technol. , 32(8), 085203. doi:10.1088/1361-6501/abff80
2021	Montoya, M., Montelongo, Y., Jiang, N., Morris, S.M., Parra-Michel, J.R. <i>Free-form laser profilometry for pipeline inspection and 3D cylindrical reconstructions.</i> IEEE Sensors J. , 21(24), 27855–27863. doi:10.1109/JSEN.2021.3130224
2020	Parra-Michel, J.R. , Gutiérrez-Hernández, D., Martínez-Peláez, R., Escobar, M.A. <i>A radial in-plane sensitivity interferometer with divergent illumination for displacement measurement.</i> Applied Sciences , 10(3), 908. doi:10.3390/app10030908
2020	Igba, V.M., Ahemen, I., et al., Parra-Michel, J.R. <i>Structural elucidation and optical properties of LiZrO₂–LiBaZrO₃ nanocomposite doped with Mn²⁺ ions.</i> Opt. Mater. Express , 10(11), 2877–2895. doi:10.1364/OME.402111